

From Animations To Science

EES 4760/5760

Agent-Based and Individual-Based Computational Modeling

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Download files

Download files for Butterfly model

Download the following files from the Download page on the class web site:

- A single ZIP file with all the files in it:
https://ees4760.jmgilligan.org/models/class_06/class_06.zip
 - Unzip after downloading.
- Or individual files listed on the download page for “Butterfly Science”:
https://ees4760.jmgilligan.org/downloads/06-butterfly_science
- Start NetLogo and load `butterfly_class_06a.nlogox`

Projects

Projects

- Team Project (Sept/Oct)
- Individual Project (whole semester)
- Detailed Assignments for Individual & Team projects are on the “[Projects](#)” page of the course web site.

Individual Semester Project

- Pick a published model to work with
 - Recommendations:
 - CoMSES OpenABM library
 - NetLogo “model library”
 - NetLogo “[modeling commons](#)”
 - *The Modeling Social Behavior* textbook
 - Or another place
- Next Tuesday (Sept. 22) we’ll discuss choosing a project model in class.
- Friday, Sept. 25: One-page description of model and thoughts for extending it
- Wed., Oct. 21: Analysis of published model.
 - Short write-up of how model works and what you saw running it
- Wed., Nov. 4: ODD for extending model
- Wed., Nov. 18: Draft of extended model
- Tue–Thur. Dec. 1–3: In-class presentations
- Sat. Dec. 19: Report (around 10 pages)

Team Project

- Team Project:
 - Each team (2–3 students) will code a model from an ODD in the textbook:
 - The *Business Investor Model* (Ch. 10–12) studies the *sensing, adaptive behavior*, and *prediction* design concepts
 - The *Telemarketer Model* (Ch. 13) studies the *interaction* design concept.
 - By Friday Sept. 18, complete the survey on Brightspace to indicate your first choice model.
 - **Optionally:** If you and 1–2 classmates want to work together as a team say who is on your team.
 - Use model to do exercises from book
 - Make presentation about what you learned (Thurs. Oct. 15).
 - Short written report Wed. Oct. 28.

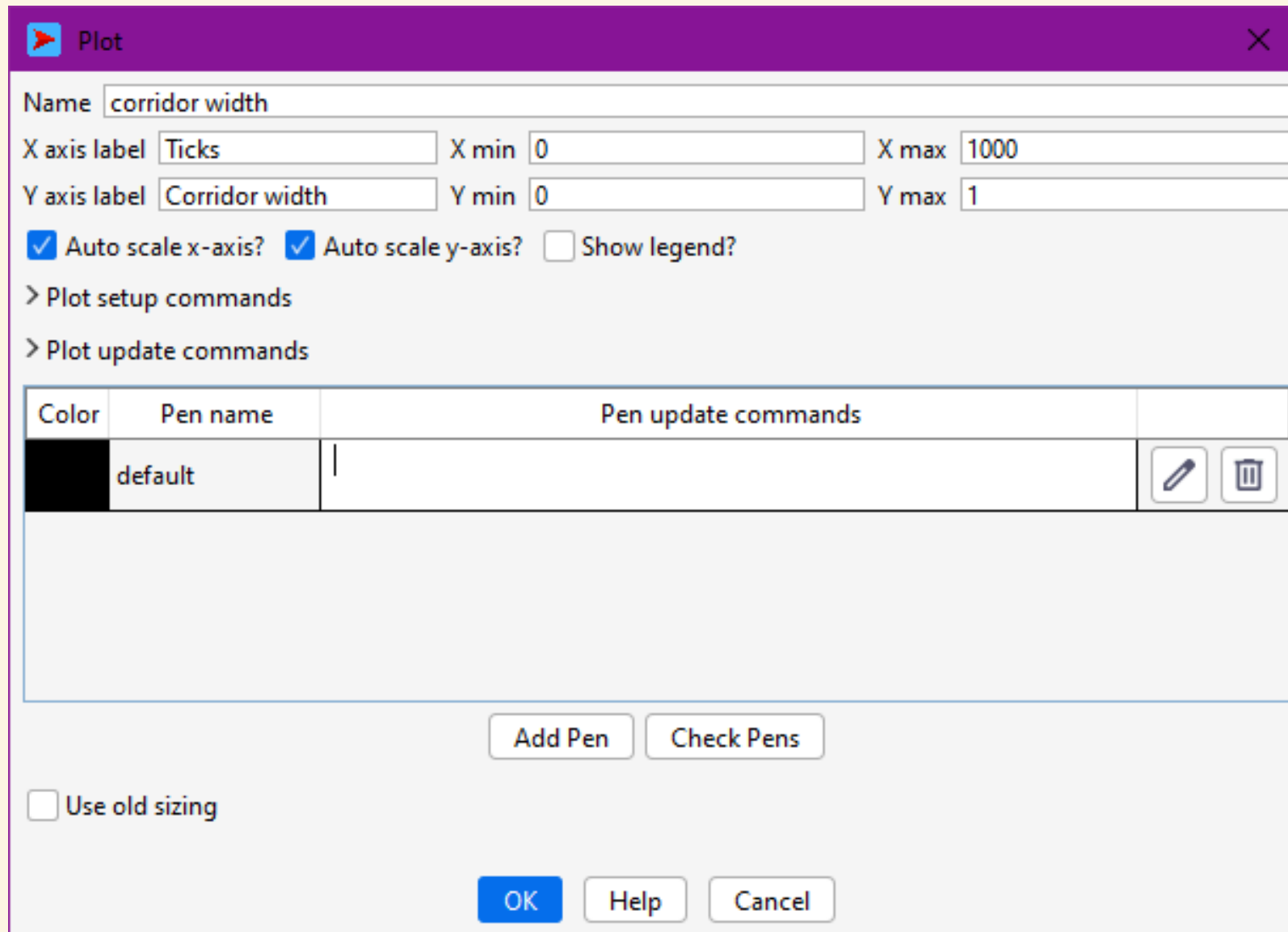
Experiments with the Butterfly Model

Experiments with the Butterfly Model

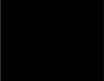
- You should have downloaded:
 - Various versions of NetLogo butterfly models.
 - A digital elevation map of real hills
https://ees4760.jmgilligan.org/models/class_06/ElevationData.txt
- Start NetLogo and load `butterfly_class_06a.nlogo`

Plot Corridor Width

- On the interface tab, add a plot



The screenshot shows a 'Plot' dialog box with a purple title bar. The 'Name' field is set to 'corridor width'. The 'X axis label' is 'Ticks', 'X min' is 0, and 'X max' is 1000. The 'Y axis label' is 'Corridor width', 'Y min' is 0, and 'Y max' is 1. There are three checkboxes: 'Auto scale x-axis?' (checked), 'Auto scale y-axis?' (checked), and 'Show legend?' (unchecked). Below these are expandable sections for 'Plot setup commands' and 'Plot update commands'. A table with three columns ('Color', 'Pen name', 'Pen update commands') contains one row with a black color swatch, the name 'default', and an empty command field. To the right of the table are edit and delete icons. At the bottom are buttons for 'Add Pen', 'Check Pens', 'Use old sizing' (unchecked), 'OK', 'Help', and 'Cancel'.

Color	Pen name	Pen update commands
	default	

- On the code tab, add a line to `go` to plot the corridor width

```
plot corridor-width
```

Enhance Interface

- Add a button to export the plot to a file:

```
export-plot "Corridor-width" (word "corridor-output-for-q-" (precision q 2) ".csv")
```

- `precision q 2` rounds `q` off to two decimal places.
- `word` combines several different things into a single text string:
“corridor-output-for-q-0.40.csv”.
- Parentheses tell NetLogo which things `word` should apply to:

```
(word a b c d ... q)
```

will combine the values of variables `a`, `b`, `c`, `d`, ..., `q` into a single text string.

Enhance Interface (cont'd.)

- Add a button to increment q by 0.1
- Add a switch `concentrate-turtles` and edit `to setup` to change the code for `crt 500` to include this:

```
ifelse concentrate-turtles
[
  setxy x0 + random 10 - 5 y0 + random 10 - 5
]
[
  setxy random-pxcor random-pycor
]
```

BehaviorSpace

- If your model is having problems, compare it to `butterfly_class_06b.nlogox`
- Open BehaviorSpace and create an experiment
 - Call it `terrain-experiment`
 - Vary `real-terrain` between `false` and `true`
 - Vary `q` from 0 to 1 in steps of 0.2
 - set `concentrate-turtles` to `true`
 - Run 20 repetitions for each value of `q`.
 - Measure `corridor-width` and `mean [elevation]` of `turtles` at the last tick only
 - Set time limit to 0 to let model run until it stops

BehaviorSpace

 Experiment 

Experiment name

Vary variables as follows (note brackets and quotation marks):

```
[ "q" [0 0.2 1]
  ["real-terrain" false true]
  ["concentrate-turtles" true]
```

Repetitions

☒ Execute combinations in sequential order

Measure runs using these reporters as metrics:

```
corridor-width
mean [elevation] of turtles
```

☐ Run metrics every step

Run metrics when

> Pre experiment commands:

Setup commands:

```
setup
```

Go commands:

```
go
```

> Stop condition:

> Post run commands:

> Post experiment commands

Time limit

OK

Help

Cancel

Run the Experiment

- Run BehaviorSpace experiment
 - Save “table” output
 - Speed things up by unchecking “Update view” and “Update plots and monitors”
- Open the analyzeBehaviorspace app at https://ees4760.jmgilligan.org/analyze_behaviorspace and use it to compare the relationship between corridor width and q for each terrain
 - If it takes too long to run, there are results in the folder “[behaviorspace-results](#)”

Practice

Practice

- Work together with a partner
- Add a button to erase the tracks of the turtles (Exercise 5.2)
- Using the realistic terrain, play with q and see what values do best at helping butterflies find mates near hilltops.

Testing

Leaving trails

Turtles leave trails with the pen (because you told them `pen-down` when you created them in `to-setup`). The pens are the same color as the turtles, so it's hard to tell the difference between turtles and trails.

It would be nice to color patches they visited yellow so you could erase the pen trails and see the red turtles contrasting with the yellow visited patches.

- Open `"butterfly_class_06b.nlogo"`
- Add code to color patches yellow when a turtle visits them:
 - At the end of `to move`, add

```
set pcolor yellow
```
- Add a button to the interface to erase the pen trails:
 - Give the button the command `clear-drawing` and the display name "erase trails"

Testing Models

- Using monitors
- Testing for consistency
 - Open `butterfly_class_06c.nlogo`
 - In `to go`, after the turtles move add:

```
if (count patches with [visited?]) != (count patches with [pcolor = yellow])  
[ print "# visited patches does not match # yellow patches." ]
```

NetLogo output

- Four commands to output to *Command Center*:
 - `show` Indicates which turtle or patch used it. Ends the line after the command.
 - `print` Ends the line after the command.
 - `type` Does not end the line.
 - `write` Does not end the line. Text strings are quoted.
- Similar commands output to the *Output* area
 - `output-show`
 - `output-print`
 - `output-type`
 - `output-write`
- See the “**Output**” section of the **Programming Guide** in the *NetLogo User Manual*.

Adding Output to Your Model

- In `to move` replace the `ifelse` block with this:

```
ifelse random-float 1 < q  
[ ; move uphill  
  let current-elevation elevation  
  move-to max-one-of neighbors [elevation]  
  if elevation < current-elevation  
  [ show "Turtle is moving downhill." ]  
]  
[ move-to one-of neighbors ] ; move randomly
```

- In `to go` replace

```
ask turtles with [not finished?] [move]
```

with

```
ask turtles [move]
```

- Set `q` to 1 on the slider
- Turtles should always move uphill. Why do they sometimes move downhill?